PROBLEM STATEMENT CODE: PS000059

SmartStreet 360: Streetlight CCMS

An IoT supervisory platform for Gandhidham Municipal Corporation that automates 3-phase fault detection, executes astronomical dimming, and slashes public lighting outages.

TEAM ID
TM-003988

TRACK CATEGORY
Pre Defined Category

INSTITUTION / ORGANIZATION
Birla Vishvakarma Mahavidyalaya (BVM), Anand, Gujarat

TEAM LEADER
Nand Patel

CHALLENGE DEFINITION

Problem Statement & Real-World Context

Focus: Root Cause & Citizen Friction

The Ground Challenge

Municipal streetlighting operations in Urban Local Bodies (like Gandhidham) rely on manual night patrols and delayed citizen complaints. Outages remain undetected for days, creating unsafe dark zones and excessive civic friction.

Indicative baseline (to be validated in pilot): 4–7 days average repair turnaround, ~35% electricity wasted from day-burning and unmanaged night burning, and recurring LED driver failures from voltage surges (>270V).

Target Beneficiaries & Flaws

- Urban Citizens & Commuters: Guaranteed illuminated roads, crime deterrence, and night safety.
- Municipal Line Supervisors: Instant geo-tagged fault work orders, eliminating night scouting.
- ULB Administration: Direct energy bill reduction and contractor SLA compliance tracking.

Existing Gap: Typical commercial smart streetlighting SCADA costs an estimated ₹15,000–₹25,000 per pole, mandates high-bandwidth 4G on every lamp, and suffers complete operational freeze if cloud connectivity drops.

SOLUTION APPROACH

Proposed Solution & Innovation Novelty

Ethos: Frugal, Accessible, Scalable

HIGH-LEVEL EXECUTIVE SUMMARY

SmartStreet 360 combines an ESP32 edge gateway with 3-phase feeder telemetry, an automated cloud rule engine (PHP/MySQL), and offline astronomical failback to automate fault detection, adaptive dimming, and rapid repair dispatch.

Core Novelty

Hybrid Feeder-to-Luminaire Architecture: monitors 3-phase feeder parameters (V, I, kW, PF) and luminaire dropouts, targeting ~95% pole-level visibility at low hardware cost.

Technical Moat

Autonomous Edge Protection: firmware trips contactors on phase loss (<160V) or surge (>270V) in <100ms and runs on offline RTC tables if the network drops.

"Seva First" Value

Grassroots Frugality & Civic Safety: targets up to ~65% lower capex (estimate vs commercial SCADA) and projected savings at ₹7.10/kWh.

PROPOSED SOLUTION DEMONSTRATION VIDEO

Video Link: https://youtu.be/smartstreet360-sfic2026-demo

End-to-end Wokwi simulation, 3-phase fault interlocks & operations dashboard

PROJECT REPOSITORY & SOURCE CODE

GitHub Link: https://github.com/shahdhairyah/SFIC_PS000059_CCMS

ESP32 C++ firmware, PHP 8 REST API, MySQL 12-table schema & web console

SYSTEM ARCHITECTURE

Technical Architecture & Methodology

End-to-End System Pipeline

01 • SENSING & ACTUATION

Feeder & Poles (Proposed Design)

• **3-Phase Electrical Bus:** Monitors R-Y-B line voltages (nominal 230V) and CT currents to identify unbalanced phases.

• **Ambient LDR Photocell:** Continuously samples ambient lux levels (<3000 ADC) for sunrise/sunset optical cross-checks.

• **Contactor Interlock:** 3-channel active-LOW optocoupled relays trigger 30A/63A heavy-duty contactors.

• **Luminaire Cluster:** Proposed pilot in Ward 3 & 7 with 70W–150W LED fixtures and 0–10V / DALI dimmable drivers.

02 • EDGE INTELLIGENCE

ESP32 RTU Gateway

• **Dual-Core FreeRTOS:** Dedicated core for high-speed ADC sampling and local fault trip interlocks (<100ms).

• **Modbus RS-485 Polling:** Reads energy meter registers for active power (kW), power factor, and cumulative kWh.

• **Hardware Safety Trip:** Trips relays immediately on undervoltage (<160V), phase drop, or surge (>270V).

• **Offline RTC Failback:** Stores astronomical sunrise/sunset in NVRAM; zero blackout during cellular outages.

03 • INGESTION & RULES

PHP 8 & MySQL Cloud

• **REST & MQTT Gateway:** Token-authenticated endpoints (/api/telemetry/feeder) ingest telemetry every 60s.

• **Message Deduplication:** Filters duplicated retransmissions using unique cryptographic message_id.

• **6-Rule Fault Engine:** Evaluates FLT_PHASE_FAIL, FLT_VOLT_SURGE, FLT_LAMP_FAIL, and DAY_BURNING.

• **12-Table Schema:** Normalized relational hierarchy: Ward → Feeder → Smart Panel → Luminaire.

04 • CIVIC SUPERVISION

Web Console & Action

• **Role-Based Access:** Dedicated operational workflows for Super Admin, Electrical Engineer, and Contractor.

• **Automated Work Orders:** Auto-dispatches repair work orders with SLA countdown deadlines & GPS tags.

• **Email & Notification Alerts:** Simulated email and in-dashboard notifications alert supervisors and linemen immediately upon a critical fault trigger.

• **Municipal Tariffs:** Ward-wise energy analytics and savings audit computed at ₹7.10/kWh commercial rate.

ARCHITECTURAL RESILIENCY

✓ **Designed in line with IS 10322 and BEE streetlighting guidelines; validated in Wokwi simulation.**

✓ **Fail-Safe Integrity:** UTC storage with Asia/Kolkata UI rendering, deduplicated message contract, and tamper-evident audit trail.

ENGINEERING STACK

Technology Stack & Feasibility

Component Readiness & Dependencies

EMBEDDED DESIGN • SIMULATION-VALIDATED (TRL-3/4)

Edge Sensing & Controller Architecture

- **Proposed MCU:** ESP32 DevKit Dual-Core 32-bit Xtensa LX6 @ 240MHz, 520KB SRAM, 4MB Flash with native Wi-Fi/BLE and 4G modem UART.
- **Proposed Sensing Circuits:** 3x AC voltage divider transformers (0–300V), 3x SCT-013 split-core CTs (0–100A), LDR ambient light sensor (<3000 dark threshold).
- **Proposed Actuation & Relays:** 3-channel optocoupled active-LOW relay modules driving 3-phase 30A industrial contactors with safety interlock.
- **Simulation Validation:** Validated on the Wokwi simulator across phase drops, voltage surges, day-burn detection, and manual maintenance switches.

CLOUD BACKEND • PHP 8 & MYSQL

High-Throughput Ingestion & Rule Engine

- **Core Architecture:** Modular PHP 8.2+ MVC design pattern, strict PDO prepared statements with parameter binding (mitigates SQL injection).
- **Database Schema:** MySQL 8.0+ / MariaDB engine featuring 12 relational tables with foreign keys & index caching (SQLite bridge for dev testing).
- **Data Contract:** Authenticated JSON telemetry contract (/api/telemetry/feeder) verifying device_id, message_id, phase voltages, and PF.
- **Security & RBAC:** Bcrypt password hashing (cost factor 12), CSRF anti-forgery tokens, secure sessions, and 4-tier granular user permissions.

OPERATIONS DASHBOARD • ZERO-BLOAT UI

Responsive Single-Page Operations Console

- **Client Architecture:** Semantic HTML5, CSS3 Grid/Flexbox design tokens, Vanilla JavaScript ES6+ fetch modules (Zero external CDN dependency).
- **Supervisory Cockpit:** Live IST clock (Asia/Kolkata), real-time phase voltage gauges, alert acknowledgement, and ward energy analytics.
- **Interactive Simulator:** Embedded in-browser test suite capable of injecting simulated phase loss, voltage surge (>270V), and lamp dropouts.
- **Field Accessibility:** Lightweight UI designed for low-bandwidth 2G/3G connectivity on lineman mobile devices.

TECHNO-ECONOMIC FEASIBILITY • BOM

Commercial Viability & Standards Compliance

- **Estimated Feeder RTU BOM:** ₹3,800–₹5,000 per panel vs ₹45,000+ (estimated) for proprietary commercial SCADA panels.
- **Zero Licensing Royalty:** Fully open-source software stack completely eliminates per-pole recurring software vendor lock-in fees.
- **Non-Invasive Retrofit (design intent):** Split-core current clamps are designed to install on existing feeder pillars without cutting power cables.
- **Standards Conformance:** Designed in line with IS 10322 lighting requirements and BEE municipal lighting benchmarks.

IMPACT ASSESSMENT

Socio-Economic Impact & Viability

Projected Outcomes (targets / estimates, to be validated in pilot)

~38%
Projected Energy Saving
Via astronomical dimming & zero day-burn

< 2 Hours
Target Fault-to-Dispatch SLA
Down from 4–7 days manual scouting

₹14.8 Lakhs
Estimated Annual Savings
Per 1,000 luminaires @ ₹7.10/kWh tariff

99.2%
Target Corridor Uptime
Ensuring safe public mobility & transit

ECONOMIC EFFICIENCY & ROI

Municipal Finance & Energy Savings

- **Commercial Tariff Optimization:** Gandhidham is projected to save ~208 MWh/year per 1,000 poles under Gujarat municipal tariff (₹7.10/kWh) through scheduled 60% stepped dimming.
- **Elimination of Day-Burning:** Automated ambient photocell logic is designed to prevent accidental daytime burning.
- **Asset Life Extension:** Instant overvoltage tripping (>270V) protects LED driver ballasts, potentially extending fixture working life.
- **Rapid Capex Payback:** Estimated payback of ~8.5–11 months from electricity bill reductions (assumption-based; to be validated in pilot).

CIVIC WELFARE & SAFETY

Public Security & Lineman Welfare

- **Dark-Zone Reduction:** Real-time pole dropout detection helps reduce hazardous dark corridors, improving safety for women, pedestrians, and motorists.
- **Proactive Civic Governance:** Automated alert ticketing targets an >80% reduction in public grievance tickets by clearing outages before citizens have to complain.
- **Lineman Occupational Safety:** Remote contactor isolation and phase trip logs reduce electrocution risk for field technicians during maintenance.
- **Proposed Pilot Corridors:** Gandhidham Port logistics axis (Ward 7) and Adipur commercial district (Ward 3).

ENVIRONMENT & NET-ZERO

Carbon Abatement & Clean Tech

- **Direct Carbon Offset:** Estimated ~170 metric tons CO2 avoided annually per 1,000 streetlights (at ~0.82 kg CO2/kWh).
- **Astronomical Night Dimming:** Dimming to 60% between 00:00–04:30 during zero-traffic windows curbs urban skyglow and preserves nocturnal ecosystems.
- **SDG Alignment:** Aligned with UN SDG 7 (Affordable & Clean Energy) and SDG 11 (Sustainable Cities & Communities).
- **Scalable Civic Model:** Provides a replicable blueprinted framework for all 150+ Urban Local Bodies across Gujarat and western India.

EXECUTION TIMELINE

Implementation Roadmap & Milestones

From Prototype to Field Rollout

PHASE 01 • COMPLETED

Simulation & Problem Study

- **Problem Study:** Analysed Gandhidham streetlighting problem (PS000059) and proposed pilot scope for Ward 3 & 7.
- **Wokwi Simulation:** Built ESP32 3-phase ADC circuit model & active-LOW relay logic in simulation.
- **Threshold Setting:** Set phase drop (<160V), surge (>270V) and daylight LDR thresholds, tested in simulation.
- **Failback Firmware:** Developed offline RTC astronomical lookup algorithm in embedded C++ (simulated).

PHASE 02 • COMPLETED

Full-Stack Web & Rule Engine

- **Relational Database:** Delivered 12-table MySQL schema with foreign keys, RBAC, and SQLite bridge.
- **Telemetry Ingestion:** Built authenticated REST API endpoints with message deduplication.
- **Automated Rules:** Engineered 6-rule fault detection engine with duplicate suppression.
- **Operator Console:** Shipped responsive HTML5/JS dashboard with live simulator and email/notification alerts.

PHASE 03 • PLANNED

Hardware-in-Loop Pilot (Proposed)

- **Pilot Commissioning:** Proposed installation of ESP32/4G CCMS panels in Ward 3 & Ward 7 corridors.
- **Cellular Telemetry:** Planned validation of 4G data streaming and offline buffering.
- **Maintenance Dispatch:** Planned field testing of email/notification alerts and lineman work-order closeout.
- **Energy Baseline:** Planned measurement of actual savings (₹7.10/kWh) against legacy baseline.

PHASE 04 • PLANNED

City-Wide Scaled Deployment

- **City-Wide Scale:** Proposed expansion to all municipal wards (scale to be confirmed with GMC).
- **Civic Integration:** API linkage with GMC Central Grievance Portal & Smart City Command Center.
- **ESCO Financing:** Proposed energy performance contracting (Energy Savings Company model).
- **ULB Handover:** Proposed lineman SOP training and institutional handover.

RISK MITIGATION MATRIX

- **Proposed: Cellular Dead-Zones:** Edge gateways designed to cache up to 7 days of telemetry locally in non-volatile flash memory, re-syncing automatically upon signal restore.
- **Proposed: Driver Incompatibility:** Stepped contactor switching provides universal dimming fallback for legacy non-dimmable LED fixtures.
- **Proposed: Saline Coastal Climate:** Sealed IP65 polycarbonate enclosures with conformal-coated industrial PCBs designed to withstand high-humidity coastal salt spray.

SQUAD CAPABILITIES

Multi-Disciplinary Team & Mentorship

Complementary Skill Sets

TL

Nand Patel

TEAM LEADER • WEB & DASHBOARD

MAIN RESPONSIBILITIES

- Website & Operations Dashboard Creation
- Real-Time Telemetry & Parameter Display
- Fault Alerting & Control UI Interface

M1

Dhyey Pagdhal

MEMBER 1 • ESP32 & WEB CONNECTIVITY

MAIN RESPONSIBILITIES

- Connecting ESP32 to Website & Server
- Telemetry Ingestion via Wi-Fi / 4G (HTTP REST)
- IoT Gateway & Cloud Data Transmission

M2

Rishi Bakrania

MEMBER 2 • HARDWARE & CIRCUITS

MAIN RESPONSIBILITIES

- Hardware Circuit Design & Prototyping
- 3-Phase Voltage & Current Sensing
- Relay Modules & Contactor Interfacing

Faculty Mentor: Dr. Ronak Vashi
Assistant Professor & SSIP Institute Coordinator, Dept. of Electronics & Communication Engg., Birla Vishvakarma Mahavidyalaya (BVM)

Ready for Jury Review & Evaluation